

Name: Ricardo Rafael Hernández Alvarenga		
School: Academica Británica Cuscatleca		
Group: 3rd hern...		Sex: Male
Date of first test: 15/11/2024	Level: 7B	Age: 7:09
Date of second test: 16/10/2025	Level: 8B	Age: 8:08

Scores

No. attempted (/46)	SAS	SAS (with 90% confidence bands)										Overall ST	PR	GR (/25)	End of KS2 indicator	Progress Category
		60	70	80	90	100	110	120	130	140						
44	107											6	68	1	107	Expected

Previous & Current Scores

Date	Age at test (yrs:mths)	Level / Form	No. attempted	SAS	SAS (with 90% confidence bands)										ST	PR	End of KS2 indicator
					60	70	80	90	100	110	120	130	140				
15/05/2025	8:03	8A	44 / 46	111											6	77	109
16/10/2025	8:08	8B	44 / 46	107											6	68	107

Analysis of Curriculum content categories

Curriculum content category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Number	30	69%	48%	21%
Measurement	9	50%	46%	4%
Geometry	3	83%	43%	40%
Statistics	4	60%	48%	12%

Analysis of Process categories

Process category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Fluency in facts and procedures	14	65%	54%	11%
Fluency in conceptual understanding	14	60%	44%	16%
Problem solving	4	80%	38%	42%
Mathematical reasoning	14	71%	47%	24%

Implications for teaching and learning

- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
 - Ricardo Rafael took *PTM7B* in November 2024 and from then until now has made expected progress in maths.
- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- Where scores are fairly evenly balanced across the curriculum categories, this suggests that Ricardo Rafael will generally demonstrate a level of understanding of mathematical concepts appropriate for this age, irrespective of whether this is in written format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Ricardo Rafael is developing the language of mathematics in line with expectations for his age group. Fluency and agility are however better developed in mental maths than in applying and understanding maths.
- Where performance across the curriculum categories are uneven, specific areas of weakness might be addressed as follows:
 - Further targeted practice in the areas identified as being relatively weaker.
 - Practical activities using equipment that is designed to help Ricardo Rafael to 'see' the thinking that lies behind any concepts that are not yet secure.
 - Get Ricardo Rafael to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Ricardo Rafael is secure in performing the basic mental calculations expected for this age group and has performed above average in this aspect. These include fluency with whole numbers and the four operations, including number facts and the concept of place value.
- Next steps should include opportunities to stretch and develop conceptual understanding, to increase mental agility and to progress the development of formal written mathematics using age-appropriate symbols and methodologies. This can be done by emphasising the higher order skills of hypothesising or predicting (What is an approximate value for 49×11 ?); designing and comparing procedures (How many different ways can we work out $53 - 18$? Which is the most efficient?); interpreting results (If $9 \times 5 = 45$, how many other calculations can we easily work out, e.g. 0.9×5 ?) and applying reasoning (If I have 39p in my purse, what coins might they be?). In the formal written development there should be an emphasis on developing the correct use of mathematical language, for example in using the '=' sign correctly as a mathematical sentence.
- By providing practice time and frequent opportunities for Ricardo Rafael to use one or more known facts to work out more facts, and by engaging Ricardo Rafael in discussion and providing opportunities to explain methods and strategies to you and his peers, Ricardo Rafael will be helped to make the connections between mental calculations and the written explanations. This will help to develop both problem solving abilities and the language of mathematics, thus improving both mental agility and the ability to plan and carry out accurate calculations in a more formal way.

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Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
MM1	Geometry	Fluency in facts and procedures	How many corners does a rectangle have?	1 / 1	60	71
MM2	Number	Fluency in facts and procedures	What is one less than thirty?	1 / 1	76	75
MM3	Number	Mathematical reasoning	What is half of ten?	1 / 1	92	93
MM4	Number	Mathematical reasoning	Write a multiple of five that is less than twenty-five.	1 / 1	52	65
MM5	Number	Fluency in conceptual understanding	Write in figures the number three hundred and seventy-two.	1 / 1	72	63
MM6	Number	Fluency in conceptual understanding	A cake is divided into four. How many pieces are there?	1 / 1	76	62
MM7	Number	Mathematical reasoning	Look at the numbers in your booklet. Write the next two numbers.	0 / 1	24	30
MM8	Number	Fluency in facts and procedures	What is three multiplied by seven?	1 / 1	20	37
MM9	Measures	Mathematical reasoning	How many five pence coins make twenty pence?	1 / 1	28	62
MM10	Geometry	Fluency in facts and procedures	How many sides does a pentagon have?	0 / 1	12	43
MM11	Measures	Fluency in facts and procedures	I have a fifty pence coin, a twenty pence coin, a five pence coin and two one pence coins. How much do I have altogether?	0 / 1	48	71
MM12	Number	Fluency in facts and procedures	Add the numbers twenty-six, eleven and thirty-four.	1 / 1	8	20
MM13	Number	Fluency in conceptual understanding	Eleven children are each given seven sweets. How many sweets are there altogether?	1 / 1	16	36
MM14	Number	Fluency in conceptual understanding	Look at the numbers in your booklet. Write down every odd number.	1 / 1	36	61
MM15	Measures	Mathematical reasoning	How many days are there in five weeks?	1 / 1	4	34
AU1a	Number	Fluency in facts and procedures	Using the three numbers below, fill in the blank spaces of three sums to make totals of 100.	0 / 1	72	65
AU1b	Number	Fluency in facts and procedures	What is the missing number?	0 / 1	48	62
AU2a-c	Number	Fluency in conceptual understanding	Lauren's house number is bigger than 25, less than 47 and an odd number.	2 / 2	24	43

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU2d	Number	Mathematical reasoning	Lauren's house number can be divided by 3. Circle Lauren's house number.	1 / 1	36	51
AU3	Number	Fluency in facts and procedures	Count the number of bricks in each pile. Circle the 2 piles that have 11 bricks in total.	1 / 1	92	84
AU4	Number	Fluency in facts and procedures	Circle the 3 piles that have 15 bricks in total.	1 / 1	84	80
AU5	Number	Fluency in facts and procedures	Here are some more bricks. They are put into 6 equal piles. How many bricks are there in each pile?	0 / 1	12	42
AU6a-c	Number	Fluency in conceptual understanding	Select the shape that is $\frac{2}{8}$ shaded. Select the shape that is $\frac{1}{2}$ shaded. Select the shape that is $\frac{1}{4}$ shaded	1 / 2	2	27
AU6d	Number	Fluency in conceptual understanding	Which 2 fractions are the same size?	0 / 1	28	38
AU7a	Measures	Fluency in conceptual understanding	How much money does she have?	0 / 1	12	25
AU7b	Measures	Mathematical reasoning	Arjun has £6.97. He buys some sweets that cost £3.42. How much money does he have left?	0 / 1	32	46
AU8	Geometry	Fluency in facts and procedures	Draw lines to join the shapes to their names.	4 / 4	15	36
AU9	Number	Mathematical reasoning	Make the addition calculation with the largest possible answer.	1 / 1	48	29
AU10	Number	Mathematical reasoning	Now make the addition calculation with the smallest possible answer.	1 / 1	44	42
AU11	Number	Problem solving	Now make the subtraction calculation with the largest possible answer.	0 / 1	24	29
AU12a	Number	Fluency in conceptual understanding	Fill in the table to show how many squares make up each L-shape.	1 / 1	48	47
AU12b	Number	Mathematical reasoning	How many squares do you need to make L-shape 4?	1 / 1	40	39
AU13	Statistics	Fluency in conceptual understanding	How many children have each number of pencils in total?	1 / 2	20	39
AU14	Number	Fluency in conceptual understanding	Here is a number spider for 24. The calculations at the end of each 'leg' should make 24. Some are wrong.	2 / 3	44	54
AU15	Measures	Mathematical reasoning	Rachel is 90 centimetres tall. Her teacher is twice as tall as she is. How tall is Rachel's teacher?	1 / 1	20	51
AU16	Measures	Problem solving	One carrot weighs 50 grams. It weighs the same as 10 raspberries. How much does 1 raspberry weigh?	1 / 1	32	49
AU17	Measures	Problem solving	One orange weighs 60 grams. One grape weighs 5 grams. How many grapes weigh the same as 1 orange?	1 / 1	32	54
AU18	Measures	Fluency in conceptual understanding	The top pencil is 5 centimetres long. How long are the other pencils?	0 / 2	38	32
AU19i	Number	Fluency in facts and procedures	Levi has some calculations to do for his homework.	1 / 1	72	64
AU19ii	Number	Fluency in conceptual understanding	Levi has some calculations to do for his homework.	0 / 1	12	32
AU19iii	Number	Fluency in facts and procedures	Levi has some calculations to do for his homework.	0 / 1	44	58
AU19iv	Number	Mathematical reasoning	Levi has some calculations to do for his homework.	0 / 1	4	28
AU20	Number	Problem solving	Nine children share a bag of sweets. Each child has 5 sweets and there are 3 sweets left. How many sweets were in the bag?	2 / 2	16	29

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU21	Statistics	Fluency in conceptual understanding	Which author is the most popular in Zain's school?	1 / 1	76	80
AU22a	Statistics	Mathematical reasoning	How many pupils chose Jeff Kinney as their favourite?	1 / 1	24	51
AU22b	Statistics	Mathematical reasoning	How many pupils took part in the survey?	0 / 1	4	31